Unified V2/V3/V4 Price Coherence: A CEX-Oracle Market-Maker with Arbitrage-Driven Convergence and LP-Fee Recycling on a Single Pool Precompile

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Abstract

On most decentralized exchanges the constant-product (V2), concentrated-liquidity (V3) and order-book/V4 venues are *separate contracts* whose prices are kept loosely aligned only by external arbitrage. We describe and measure a different design realized on the Lux DEX: V2, V3 and V4 are three *liquidity modes of a single pool* hosted by one EVM precompile (`0x9999`), sharing one pool identity, one fee tier and one maker peg. A single market-maker tracks a live centralized-exchange (CEX) mid (Kraken public market data) and pegs the book to it; an arbitrage loop harvests the residual requote-lag gap, pulling the on-chain price back to the reference. Because the LP is also the arber on a sovereign L1, only the EIP-1559 base fee is a true cost, so the loop recycles fee income to the dominant LP net of a small burn. We implement the CEX oracle and the coherence/arbitrage harness, and drive real fills on the Lux devnet `0x9999 PoolManager` (chain `96367`). We report exact on-chain costs—`swap` 73,608 gas, a maker requote (`deposit`+`place`) 114,272 gas, 25 gwei base fee—and establish the CLOB fee mechanism empirically (a parity fill of size 10 returns exactly 10: the V4 CLOB charges *no* taker fee; the LP earns the quoted spread, while the 30 bps tier applies to the V2/V3 curve). A simulation calibrated by these measurements and driven by a real 12-hour Kraken ETH/USD trace (7.66 bps per-minute volatility) shows the three modes track the oracle within 0.3–13 bps for requote intervals of 2 s–10 min, with cross-version dispersion below 14 bps—inside the 30 bps no-arbitrage band. The cycle is self-sustaining at every interval; the burn is minimized ($\sim$ \$100–160/day for three pools) at a 60–300 s requote band where requote-gas and arbitrage-gas balance, with a break-even external volume of $\sim$ \$33–53k/day. We give the no-arbitrage-band model relating requote interval, volatility, fee and gas to tracking error and break-even volume, and discuss oracle-manipulation, MEV and inventory risks.

1 Introduction

A decentralized exchange that lists an asset also traded on centralized venues faces a standing problem: its on-chain price must stay aligned with the broader market, or it becomes a faucet for arbitrageurs at the liquidity providers' expense. The conventional remedy is passive—deploy automated market-maker (AMM) pools and let external arbitrage drag the price toward the global mid. This is robust but leaks value to outside arbitrageurs and provides no guarantee of *coherence* across the several pool types (V2 constant-product, V3 concentrated liquidity, V4/order-book) a mature DEX runs in parallel, each a separate contract with its own price.

This paper studies an *active* design realized on the Lux DEX, where the three "versions" are not separate contracts but *three liquidity modes of one pool* inside a single EVM precompile. A single market-maker, which we operate as the dominant LP, pegs the shared book to a live CEX reference; an arbitrage loop, which we also operate, harvests the residual gap that opens during requote lag and pulls the price back. Because on a sovereign L1 the operator is also the validator, the priority tip recycles and only the base fee is burned, so the loop recycles fee income to the LP net of a small, measurable burn.

Contributions.

1. **A unified-pool coherence mechanism** (§2,§3): pegging one maker quote to a CEX oracle keeps V2/V3/V4 coherent *by construction*, because they share one pool identity and one peg—not three reconciled AMMs.

2. **An implementation** (§4): a Kraken public-data oracle as a drop-in maker price source, and a coherence/arbitrage harness driving the `0x9999` money path; both build and ship with tests, and the oracle is live-proven against the real Kraken API.

3. **Real on-chain measurements** (§5) on Lux devnet: exact per-operation gas, block cadence, latency, and an empirical determination of the CLOB fee mechanism (no taker fee; LP earns the spread).

4. **A calibrated study** (§5,§6): a simulation pinned to the measured gas/fee/block parameters and driven by a real Kraken trace, quantifying tracking error, cross-version dispersion, and the fee/burn economics, and the no-arbitrage-band model that explains them (§3).

We are careful throughout to label every quantity $^{\mathsf{dev}}$ (real, from an accepted devnet receipt or log) or $^{\mathsf{sim}}$ (from the calibrated model). No figure is a vendor number.

2 The Lux unified DEX

The Lux DEX money path is a single EVM precompile at `0x9999`, the V4-style `PoolManager`. A pool is identified by a `PoolKey` (currency pair, fee tier, tick spacing, hooks). The precompile hosts three liquidity modes against the same pool identity:

"Version"	Mode on `0x9999`	Mechanism	Devnet status
V4	native CLOB	`swapPlace`/`swapDeposit`/`swap`	live, fills proven
V3	concentrated liquidity	`modifyLiquidity` (sqrtPrice/ticks)	precompile present
V2	constant product $xy{=}k$	`BindAMMPool`→router blend	present; no bind selector

Read-only views are separate precompiles, all live on devnet: `0x9998` (Quoter, advisory swap quote), `0x9997` (StateView, pool/book/position read), and `0x9012` (LXRouter, which blends the V4 native pool, the V2 constant-product reserves, and external venues). The router's V2 quote reads a reserve row bound in `0x9999` storage; its "external V3" path is a stub, because the *native* V3 equivalent is `0x9999` concentrated liquidity, not a foreign Uniswap-V3 deployment.

Remark 2.1 (Coherence is structural)**.** *Because the three modes share one `PoolKey` identity and one fee tier, a market maker that pegs* one *quote pegs all three. There is no cross-contract reconciliation: the "unified V2/V3/V4 coherence" reduces to a single price the maker controls. This is the central architectural advantage over a three-contract Uniswap deployment.*

3 Mechanism

Setup. Let $P(t)$ be the live CEX reference mid (Kraken). The maker holds a peg P_{peg}, refreshed to $P(t)$ every requote interval T. The pool's marginal price $m(t)$ starts at the peg and is moved by trades. In the model the operator is the dominant LP and runs the arbitrage loop; external flow is exogenous.

No-arbitrage band. An arbitrageur closing a price gap $\delta = (m - P)/P$ trades size s (base units) at price P. The round trip earns $\delta\, sP$, pays the fee $f\, sP$ (the 30 bps pool tier on V2/V3, or the maker's quoted spread on V4), and burns gas worth $g = \gamma\, \beta_{\text{base}}\, p_{\text{LUX}}$ in quote units, where γ is the swap gas, β_{base} the base fee, and p_{LUX} the LUX price. It profits iff

$$|\delta| \;>\; \beta \;\equiv\; f \;+\; \frac{g}{sP}. \tag{1}$$

β is the half-width of the no-arbitrage band: the pool price is pinned to P within $\pm\beta$ whenever the arb is active. On a sovereign L1 the tip recycles to the operator's validator, so g is the base-fee burn only and $g/(sP) \ll f$; hence $\beta \approx f$ (fee-dominated).

Tracking error and the requote/arb trade-off. Between requotes the reference drifts while the maker quote is stale; the taker-facing tracking error grows as $\mathbb{E}|m - P|/P \approx \sigma\sqrt{T}$ for per-unit-time volatility σ, until a requote ($P_{\text{peg}} \leftarrow P$, gas γ_{rq} per pool) or an arb (when the drift exceeds β) resets it. Thus two controls hold coherence: *fast requoting* (tight tracking, high requote-gas) and *arbitrage* (caps drift at β, one swap per correction). The optimum minimizes total burn for a tolerated tracking error.

Cycle economics. When the LP and the arber are the same operator, arb fees and arb adverse selection are internal washes. Real income is the fee on *external* volume V_{ext}; real cost is the base-fee burn of requoting and of the operator's arb swaps:

$$\text{net} \;=\; f\,V_{\text{ext}} \;-\; p_{\text{LUX}}\,\beta_{\text{base}}\Big(N_{rq}\,\gamma_{rq} + N_{arb}\,\gamma_{arb}\Big), \tag{2}$$

with $N_{rq} = \#\text{pools} \cdot \lfloor \mathcal{H}/T \rfloor$ requotes over horizon $\mathcal{H}$ and N_{arb} arb swaps. The cycle is self-sustaining (net > 0) above the break-even external volume

$$V^{\star} \;=\; \frac{p_{\text{LUX}}\,\beta_{\text{base}}\,(N_{rq}\gamma_{rq} + N_{arb}\gamma_{arb})}{f}. \tag{3}$$

4 Implementation

Kraken oracle. We add a price source reading Kraken's public, unauthenticated market-data endpoints (`/0/public/Ticker` for BBO, `/0/public/Depth` for book). In the maker it is an aggregator plug-in (the same shape as the existing on-chain oracle sources) exposed as the live mid the seeder tracks; a composition primitive lets the CEX feed (listed majors) and the native-AMM feed (LUX/$*$) coexist behind the single feed the seeder consumes. In the arbitrage client it is a scalar-mid reader. The oracle is hermetically tested and live-proven (a real `BTC/USD` mid of $59,507.75 pulled from the public API during testing).

Coherence harness. The harness drives the full loop on `0x9999` using only the proven primitives (`Initialize`, `Deposit`, `Place`, `Swap`, and the `DEXFill` log reader). Each round pegs the maker bid to the oracle mid, has the taker trade it, and records a per-round sample—realized fill price, implied fee, gas, block, latency—so the economics are *measured*. A drift-injection window mispriced relative to the reference exercises the arbitrage leg.

5 Methodology and results

On-chain [dev]. We ran the harness against Lux devnet (chain `96367`, `0x9999`, the genesis-funded index-0 account). Gas is exact from receipts; block timing from block timestamps; fills from `DEXFill` logs.

operation	gas ($^{\text{dev}}$)	base-fee burn @25 gwei
`swapDeposit`	41,584	0.001040 LUX
`swapPlace`	72,688	0.001817 LUX
`swap` (taker/arb)	73,608	0.001840 LUX
maker requote (`deposit`+`place`)	114,272	0.002857 LUX

The devnet builds one transaction per block; warm inter-block time is 2–5 s, with 25–50 s gaps after the builder idles. Base fee is pinned at the 25 gwei floor. Taker-swap latency was $\approx$5 s in steady state with cold-start spikes to 40–67 s. Six `DEXFill`s settled to accepted state.

Proposition 5.1 (CLOB fee mechanism, empirical)**.** *The V4 CLOB charges no taker fee; the LP's CLOB income is the quoted spread. A taker selling* 10 *base units at parity into a parity bid received exactly* 10 *(block 14), i.e. zero fee. The* 30 *bps* `Fee` *tier applies to the V2/V3 curve, not to CLOB matches.*

Simulation [sim]. The devnet block-production cadence makes long horizons impractical to run live, so we drive a calibrated model: V2 ($xy{=}k$), V3 (concentrated, 20× depth), V4 (CLOB, 5× depth, 1 bps tick) tracking the *real* Kraken ETH/USD 1-minute trace (721 points, 12 h, $\sigma_{1m} = 7.66$ bps, 221 bps realized range). Calibration is the measured $\gamma_{arb} = 73{,}608$, $\gamma_{rq} = 114{,}272$, $\beta_{\text{base}} = 25$ gwei, block = 2 s, with $f = 30$ bps and $p_{\text{LUX}} = \$12.5$; pool depth, concentration and organic-flow rate are explicit modelling assumptions. Table 1 sweeps the requote interval.

Table 1: Requote-interval sweep [sim], $f{=}30$ bps, calibrated by §5. RMS tracking error per mode, p99 cross-version dispersion, arb count, external fee income and base-fee burn over the 12 h horizon (three pools), and net.

T (s)	V2 RMS (bps)	V3 RMS (bps)	V4 RMS (bps)	disp. p99 (bps)	arbs	fee ($)	burn (LUX)	net ($)
2	0.69	0.26	0.29	1.90	0	15,370	185.1	13,056
6	1.06	0.55	0.58	1.90	0	15,370	61.7	14,598
12	1.56	1.00	1.03	3.80	0	15,370	30.9	14,984
30	2.91	2.33	2.35	3.80	0	15,370	12.3	15,216
60	5.05	4.49	4.52	5.70	88	15,370	6.3	15,291
120	7.27	6.71	6.73	7.60	522	15,370	4.1	15,319
300	11.21	10.42	10.46	11.40	1,453	15,370	3.9	**15,321**
600	13.40	12.82	12.83	13.31	2,633	15,370	5.5	15,302

Four observations. (i) Tracking error grows monotonically with T, from ~ 0.3 bps at 2 s to ~ 13 bps at 10 min, as Eq. (1)–model predicts. (ii) V3 (concentrated) tracks tightest and V2 (shallow) loosest, the spread being small (~ 0.3–0.6 bps): deeper liquidity resists organic-flow impact. (iii) Cross-version dispersion stays well inside the 30 bps band ($1.9 \to 13.3$ bps p99): coherence holds because one maker pegs all three. (iv) Arbitrage fires only once T is large enough that drift exceeds β (≥ 60 s here), at which point it—not requoting—is the corrective force.

6 Economics: is the cycle self-sustaining?

By Eq. (2), income is the external fee and cost is the base-fee burn of requoting plus arbitrage. The two burn terms trade off: requote burn dominates at small T (185 LUX at 2 s), arb burn at large T; total burn is minimized in a 60–300 s band at ∼4–6 LUX over 12 h (∼$100–160/day for three pools). Against the modelled external fee income (∼ $30k/day) this is a 200–300× margin, and net is positive at *every* interval. By Eq. (3) the break-even external volume falls from $257k/day (2 s) to ∼$33k/day in the optimal band.

Proposition 6.1 (Self-sustaining)**.** *Under the calibrated model the cycle is self-sustaining for any external daily volume above* ∼$50*k; in the burn-optimal* 60–300 *s requote band the break-even is* ∼$33–53*k/day.*

The policy implication is to *requote at* ∼60–120 *s and let the arbitrage loop hold the no-arbitrage band*, rather than pay for fast requoting: an arb correction is one 73,608-gas swap, cheaper than re-pegging three pools every few seconds.

7 Related work

Constant-product and concentrated-liquidity AMMs [1, 2] and the singleton V4 `PoolManager` [3] define the venue surface; our pools are V4-style but host the order book, concentrated liquidity and a constant-product reserve as modes of one identity. Loss-versus-rebalancing [4] formalizes the adverse selection an oracle-lagging LP suffers; our no-arbitrage band (Eq. (1)) and self-arb policy are the operational response, and our requote sweep quantifies the LVR/gas trade-off directly. Oracle-pegged or "oracle AMM" designs [5] reduce arbitrage leakage by pricing off an external feed; we go further by making the operator both the dominant LP and the arber on a sovereign L1, so the leakage is recaptured net of base-fee burn.

8 Limitations and future work

The on-chain horizon was short: the devnet builder stalls between transactions (a known node-version regression), so the long-horizon tracking and economics are from the calibrated model, not a multi-hour live run; the real measurements pin the model's cost parameters and its fee mechanism (Prop. 5.1). The native V2 mode has no external bind selector today—`BindAMMPool` is only invoked internally after a fill—so a permissioned operator-only `bindAMMPool` entry point is needed to seed real V2 reserves; V3/V4 are ready on `0x9999`. The model's pool depth, concentration factor and organic-flow rate are explicit assumptions; tightening them with live order-flow data is future work, as is multi-CEX median pricing with staleness and deviation circuit-breakers (§below).

Risks. The peg trusts the CEX feed: a spoofed or halted feed mis-pegs the book (mitigation: median across feeds, staleness/deviation breakers, per-requote move clamps). Requote-lag quotes are picked off by external arbitrageurs—this *is* the tracking error—trading off against requote gas. A trending market fills one side and accumulates inventory, handled by inventory-skew quoting and size limits. A single dominant maker pegged to one CEX is a centralization point appropriate for bootstrapping a market, not an end state.

9 Conclusion

Hosting V2, V3 and V4 as liquidity modes of one pool precompile turns "cross-version price coherence" into a single maker peg, which a CEX oracle drives and an arbitrage loop defends within a fee-width no-arbitrage band. On Lux devnet we measured the exact costs of the loop and established that the V4 CLOB earns the spread rather than a fee; a calibrated model driven by a real Kraken trace shows the three modes tracking the oracle within 0.3–13 bps and staying coherent within 14 bps, with the fee-recycling cycle self-sustaining above a modest external volume. The remaining gaps—a block-production stall and a missing V2 bind selector—are operational, not architectural.

References

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